

Subject Description Form

Subject Code	ME5302
Subject Title	Principles of Robotic Mechanisms
Credit Value	3
Level	5
Pre-requisite/ Co-requisite/ Exclusion	Nil.
Objectives	1. To equip students with a comprehensive understanding of the theoretical principles that enable to study and to develop advanced robotic systems. 2. To provide students with advanced methodologies to design, predict, and simulate the behaviour of robots and other autonomous systems.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a) Mathematically model and analyse the mechanical properties of complex robotic systems. b) Design robotic systems to perform a specific task while satisfying various constraints. c) Understand the main feedback paradigms that are used to command the motion of robots. d) Numerically simulate the motion and behaviour of robotic mechanisms.
Subject Synopsis/ Indicative Syllabus	Kinematic Modelling. Degrees-of-freedom, rigid body transformations, serial and parallel mechanisms, forward kinematics, DH formulation, inverse kinematics, differential kinematics, omnidirectional and nonholonomic systems, kinematics of complex robotic systems. Dynamic Modelling. State-space representations, mass distribution of mechanical linkages, inertia tensors, Newton-Euler formulation, Lagrangian dynamics, Hamiltonian dynamics, Cartesian-space dynamics, dynamics of non-inertial mobile robots, numerical integration of second-order equations. Feedback-Based Methods. Principles of feedback actions, closed-loop models, equivalent dynamics, force/impedance control, virtual stiffness, parameter identification. Robot Simulation. Common simulation tools for robot mechanisms, kinematic/dynamic simulation, 3D visualisers, Matlab/Python for robotics, cases of study, group projects.

Teaching/Learning Methodology	Lectures will be used to provide students with theoretical knowledge required for an advanced in-depth study of modern robotic systems. (learning outcomes a to c)					
	Assignments and group projects will be used to provide students with a hands-on experience to solve practical problems using the acquired knowledge. (learning outcome a to d)					
Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)			
			a	b	c	d
	1. Assignments	30%	✓	✓	✓	✓
	2. Project	20%	✓	✓	✓	✓
	3. Class test	10%	✓	✓	✓	
	3. Final exam	40%	✓	✓	✓	
	Total	100 %				
	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:					
	Overall Assessment: 0.40 x End of subject exam + 0.60 x Continuous assessment.					
	1. The continuous assessment aims at evaluating the progress of the students’ study, assisting them in self-monitoring the respective learning outcomes, and applying the knowledge learnt in practical situations.					
2. The examination is used to assess the knowledge acquired by the students for understanding and analysing the problems critically and independently; as well as to determine the degree of achieving the subject learning outcomes						
Student Study Effort Expected	Class contact:					
	▪ Lecturers				33 Hrs.	
	▪ Tutorials / Exercises / Laboratory				6 Hrs.	
	Other student study effort:					
	▪ Self-learning				36 Hrs.	
	▪ Assignments and project				40 Hrs.	
	Total student study effort				115 Hrs.	

Reading List and References	1. M. W. Spong S. Hutchinson, and M. Vidyasagar, Robot Modeling and Control, Wiley, latest edition. 2. K. Lynch, Modern Robotics: Mechanics, Planning, and Control, Cambridge University Press, latest edition. 3. B. Siciliano, L. Sciavicco, L. Villani, and G. Oriolo, Robotics, Modelling, Planning and Control, Springer, latest edition.
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